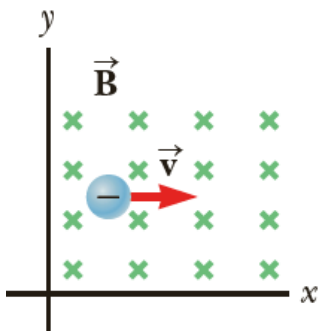




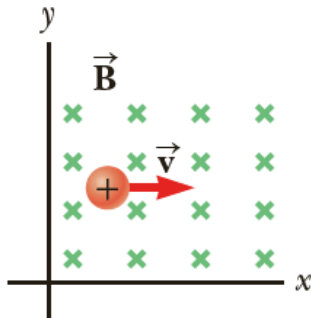
Directions: Show all steps required to arrive at your answer and give answers with the correct units if applicable. **A correct answer with no justification will receive no credit.**

1. Indicate the direction the particle will move.

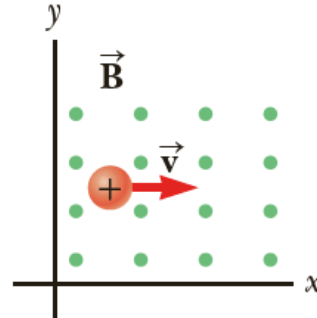
a) _____



b) _____

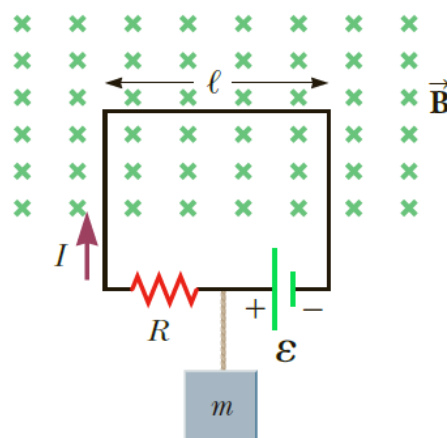


c) _____

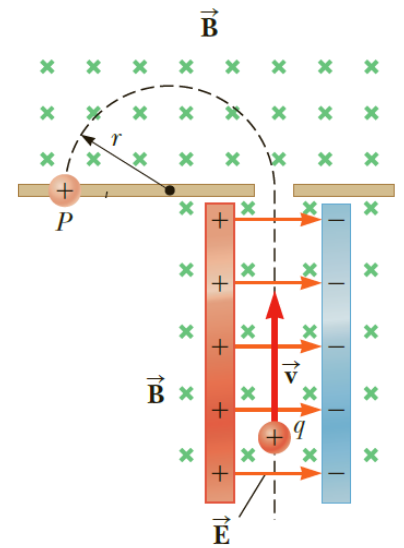


2. A particle (mass = .005 kg, charge = $2.0 \mu\text{C}$) moves in the positive direction along the z axis with a velocity of 50000.0 m/s . It enters a magnetic field of $\vec{B} = (1.0\mathbf{i} - 3.0\mathbf{j} + 7.0\mathbf{k}) \text{ T}$. What is the magnitude of the acceleration of the particle?

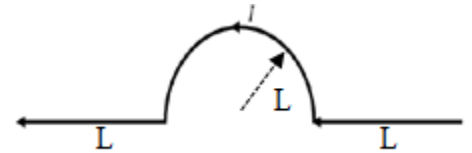
3. Mass $m = 1.0 \text{ kg}$ is suspended vertically at rest by an insulating string connected to a circuit partially immersed in a magnetic field as shown. The magnetic field has magnitude $B = 2.0 \text{ T}$ and the length $l = 0.5 \text{ m}$. If the battery voltage is $\mathcal{E} = 120 \text{ V}$, find the required value of resistance R .



4. Brood about the mass spectrometer shown schematically. The electric field between the parallel plates is 950 V/m , and the magnetic fields in all regions have magnitude of 0.930 T . Calculate the radius of the path in the system for a charged ion with mass $2.2 \times 10^{-26} \text{ kg}$.



5. A section as shown carries a current of $I = 3.0$. Wow, that's a lot of current. The wire has straight segments of length $L = 0.2 \text{ m}$ as shown and the circular section has a radius of $L = 0.2$ as well. Calculate the magnetic field at the center of the circular section.



6. The two wires shown have opposite currents, I and $2I$, and are separated by a distance r . If the wires have equal lengths L , find the direction and magnitude on the wire with current $2I$.

